

Material Safety Data Sheet



ECO-STAR PRO CHLOR

Section 1. Chemical product and company identification

Trade name : ECO-STAR PRO CHLOR
Product use : Bleach
Supplier :
Ecolab Inc. International Division
370 N. Wabasha Street
St. Paul, MN 55102
Code : 900043-95
Date of issue : 25-April-2008

EMERGENCY HEALTH INFORMATION: 1-800-328-0026
Outside United States and Canada CALL 1-651-222-5352 (in USA)

Section 2. Composition, information on ingredients

United States

<u>Name</u>	<u>CAS number</u>	<u>% by weight</u>
sodium hypochlorite	7681-52-9	8.4

Section 3. Hazards identification

Physical state : Liquid. [Liquid.]
Emergency overview : DANGER !

CAUSES EYE BURNS.
CAUSES SEVERE SKIN IRRITATION. Untreated contact may cause severe irritation or chemical burns.
May cause burns to mouth, throat and stomach.
CAUSES RESPIRATORY TRACT IRRITATION.
Do not ingest. Do not get in eyes, on skin or on clothing. Avoid breathing vapors, spray or mists. Use only with adequate ventilation. Keep container closed. Wash thoroughly after handling.

Potential acute health effects

Eyes : Corrosive to eyes.
Skin : Severely irritating to the skin. Untreated contact may cause severe irritation or chemical burns.
Inhalation : Irritating to respiratory system.
Ingestion : May cause burns to mouth, throat and stomach.

See toxicological information (section 11)

Section 4. First aid measures

Eye contact : In case of contact, immediately flush eyes with cool running water. Remove contact lenses and continue flushing with plenty of water for at least 15 minutes. Get medical attention immediately.

Skin contact : In case of contact, immediately flush skin with plenty of water for at least 15 minutes while removing contaminated clothing and shoes. Get medical attention immediately. Wash clothing before reuse. Clean shoes thoroughly before reuse.

Inhalation : If inhaled, remove to fresh air. If exposed person is not breathing, give artificial respiration or oxygen applied by trained personnel. Get medical attention if irritation persists.

Ingestion : If material has been swallowed and the exposed person is conscious, give small quantities of water to drink. Do not induce vomiting. Never give anything by mouth to an unconscious person. Get medical attention immediately.

Section 5. Fire fighting measures

- Flash point** : > 100°C
- Hazardous thermal decomposition products** : Decomposition products may include the following materials:
halogenated compounds
metal oxide/oxides
- Fire-fighting media and instructions** : Use an extinguishing agent suitable for the surrounding fire.
- Dike area of fire to prevent runoff.
- In a fire or if heated, a pressure increase will occur and the container may burst.
- Special protective equipment for fire-fighters** : Fire-fighters should wear appropriate protective equipment and self-contained breathing apparatus (SCBA) with a full face-piece operated in positive pressure mode.

Section 6. Accidental release measures

- Personal precautions** : Immediately contact emergency personnel. Stop leak if without risk. Use suitable protective equipment. Keep unnecessary personnel away. Do not touch or walk through spilled material.
- Environmental precautions** : Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers. Inform the relevant authorities if the product has caused environmental pollution (sewers, waterways, soil or air).
- Methods for cleaning up** : If emergency personnel are unavailable, contain spilled material. For small spills, add absorbent (soil may be used in the absence of other suitable materials), scoop up material and place in a sealable, liquid-proof container for disposal. For large spills, dike spilled material or otherwise contain it to ensure runoff does not reach a waterway. Place spilled material in an appropriate container for disposal.

Section 7. Handling and storage

- Handling** : Do not ingest. Do not get in eyes, on skin, or on clothing. Avoid breathing vapors, spray or mists. Keep container closed. Use only with adequate ventilation. Wash thoroughly after handling.
- Storage** : Keep out of reach of children. Keep container in a cool, well-ventilated area. Keep container tightly closed.
- Do not store above the following temperature: 40°C

Section 8. Exposure controls, personal protection

- Engineering measures** : Use only with adequate ventilation. If user operations generate dust, fumes, gas, vapor or mist, use process enclosures, local exhaust ventilation or other engineering controls to keep worker exposure to airborne contaminants below any recommended or statutory limits. Provide suitable facilities for quick drenching or flushing of the eyes and body in case of contact or splash hazard.

Personal protection:

- Eyes** : Use chemical splash goggles. For continued or severe exposure wear a face shield over the goggles.
- Hands** : Use chemical-resistant, impervious gloves.
- Skin** : Wear suitable protective clothing.
- Respiratory** : Avoid breathing vapors, spray or mists.

Name

chlorine

Exposure limits

ACGIH TLV (United States, 1/2004).

STEL: 2.9 mg/m³ 15 minute(s). Form: All forms

STEL: 1 ppm 15 minute(s). Form: All forms

TWA: 1.5 mg/m³ 8 hour(s). Form: All forms

TWA: 0.5 ppm 8 hour(s). Form: All forms

OSHA PEL (United States, 8/1997).

CEIL: 3 mg/m³ Form: All forms

CEIL: 1 ppm Form: All forms

Section 9. Physical and chemical properties

Physical state	: Liquid. [Liquid.]
Color	: Yellow. [Light]
Odor	: chlorine
pH	: 12.5 [Conc. (% w/w): 100%]
Specific gravity	: 1.154

Section 10. Stability and reactivity

Stability	: The product is stable. Under normal conditions of storage and use, hazardous polymerization will not occur.
Reactivity	: Extremely reactive or incompatible with acids. Slightly reactive to reactive with metals. Mixing this product with acid or ammonia releases chlorine gas.
Hazardous decomposition products	: Under normal conditions of storage and use, hazardous decomposition products should not be produced.

Section 11. Toxicological information

Potential acute health effects

Eyes	: Corrosive to eyes.
Skin	: Severely irritating to the skin. Untreated contact may cause severe irritation or chemical burns.
Inhalation	: Irritating to respiratory system.
Ingestion	: May cause burns to mouth, throat and stomach.

Potential chronic health effects

Target organs	: Contains material which may cause damage to the following organs: upper respiratory tract.
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Section 12. Ecological information

Section 13. Disposal considerations

Waste disposal	: The generation of waste should be avoided or minimized wherever possible. Avoid dispersal of spilled material and runoff and contact with soil, waterways, drains and sewers. Disposal of this product, solutions and any by-products should at all times comply with the requirements of environmental protection and waste disposal legislation and any regional local authority requirements.
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Consult your local or regional authorities.

Section 14. Transport information

Regulatory information	UN number	Proper shipping name	Class	Packing group	Additional information
UN Classification	UN1791	HYPOCHLORITE SOLUTION	8	III	-

APPLIES ONLY DURING ROAD TRANSPORT

Any variation of the shipping description based on the packaging is not addressed.

Section 15. Regulatory information

HCS Classification : Corrosive material
Target organ effects

Section 16. Other information

Hazardous Material Information System (U.S.A.) :	Health	*	3
	Flammability		0
	Physical hazards		0

Date of issue : 25-April-2008.
Responsible name : Regulatory Affairs
Date of previous issue : No previous validation.

Notice to reader

The above information is believed to be correct with respect to the formula used to manufacture the product in the country of origin. As data, standards, and regulations change, and conditions of use and handling are beyond our control, NO WARRANTY, EXPRESS OR IMPLIED, IS MADE AS TO THE COMPLETENESS OR CONTINUING ACCURACY OF THIS INFORMATION.